INCH-POUND

MIL-DTL-83513/9E
25 June 2002
SUPERSEDING
MIL-PRF-83513/9D
15 August 1997

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, SOCKET CONTACTS, CLASS P, CRIMP TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83513.

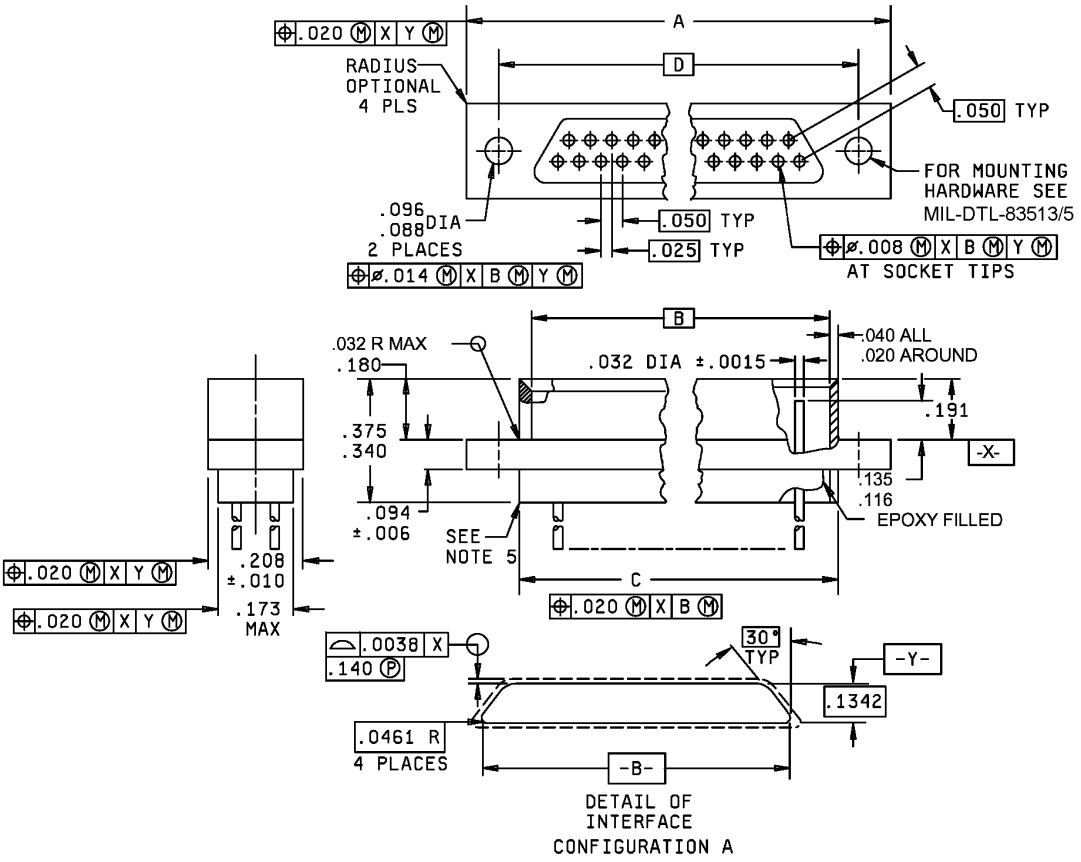


FIGURE 1. Connector, receptacle, .050 spacing.

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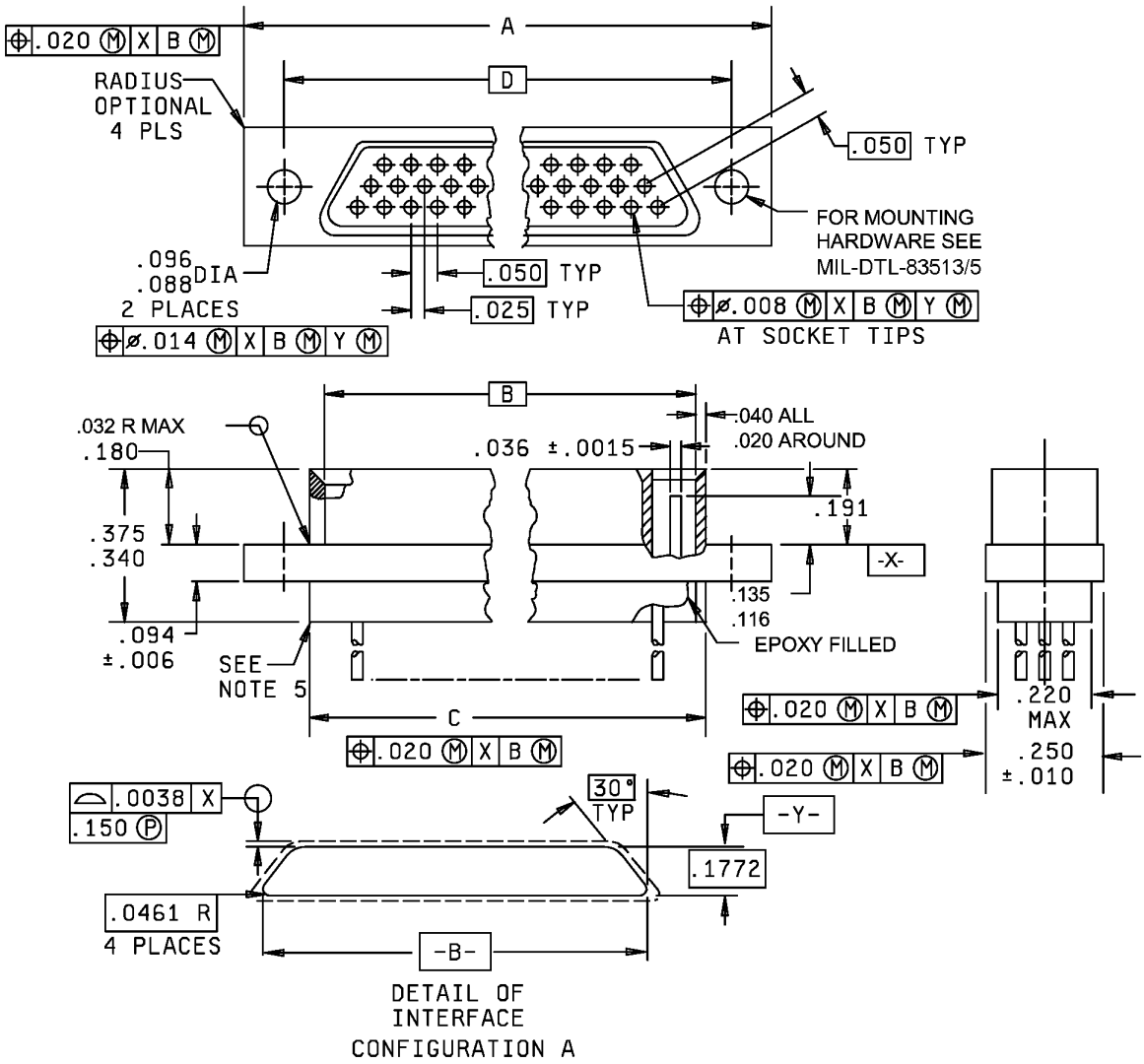


FIGURE 1 Connector, receptacle .050 spacing - Continued.

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Number of cavities	A ±.010	B BSC	C +.010 -.018	D BSC
9	.778 (19.76)	.2922 (7.422)	.398 (10.11)	.565 (14.35)
15	.928 (23.57)	.4422 (11.232)	.548 (13.92)	.715 (18.16)
21	1.078 (27.38)	.5922 (15.042)	.698 (17.73)	.865 (21.97)
25	1.178 (29.92)	.6922 (17.582)	.798 (20.27)	.965 (24.51)
31	1.328 (33.73)	.8422 (21.392)	.948 (24.08)	1.115 (28.32)
37	1.478 (37.54)	.9922 (25.202)	1.098 (27.89)	1.265 (32.13)
51	1.428 (36.27)	.9422 (23.932)	1.048 (26.62)	1.215 (30.86)

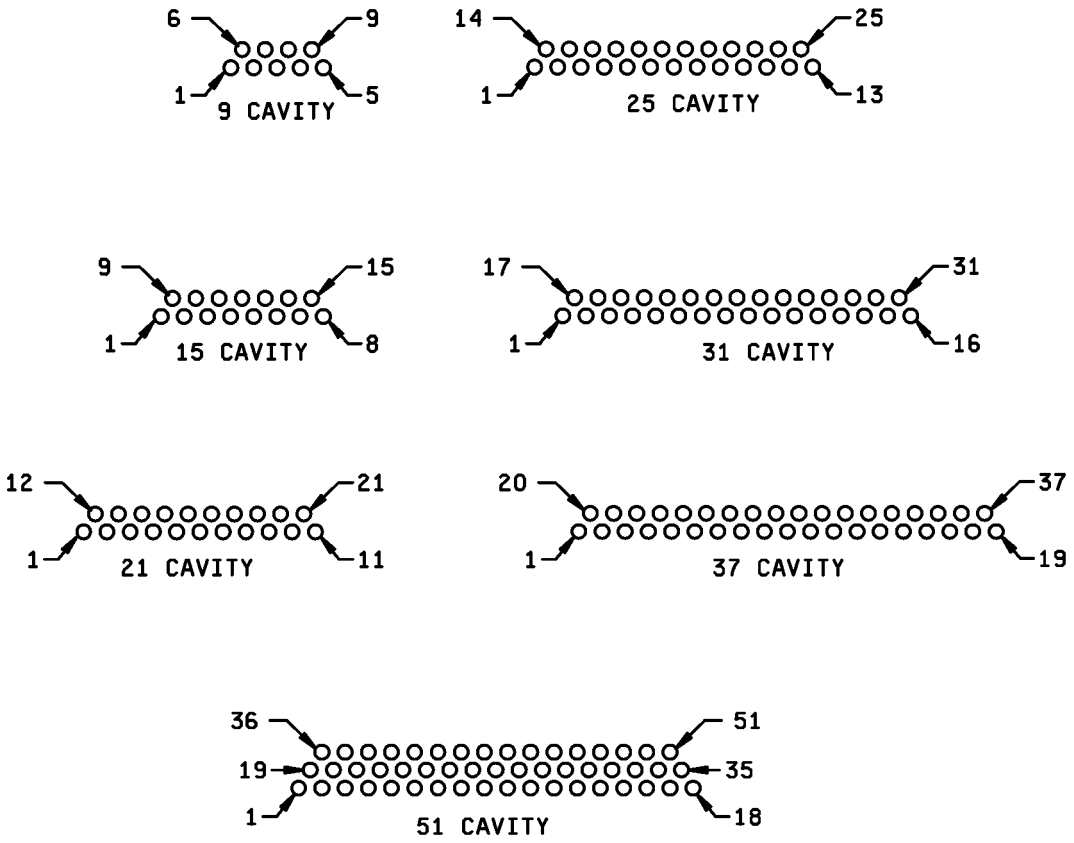
Inches	mm
.0015	0.038
.0019	0.048
.002	0.05
.0038	0.097
.006	0.15
.008	0.20
.010	0.25
.014	0.36
.020	0.51
.025	0.64
.027	0.69
.032	0.81
.040	1.02
.0461	1.171
.047	1.19
.050	1.27
.088	2.24
.0886	2.250
.093	2.36
.094	2.39
.096	2.44
.116	2.95
.135	3.43
.137	3.48
.173	4.39
.1772	4.501
.180	4.57
.208	5.28
.220	5.59
.250	6.35
.340	8.64
.375	9.52

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Metric equivalents are in parentheses or columns.
5. Plane from which the lead length is measured.

FIGURE 1. Connector, receptacle, .050 spacing - Continued.

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NOTES:
1. Engaging face of socket insert shown.
2. Cavity identification numbers are for reference only and do not appear on part.

FIGURE 2. Insert arrangement.

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REQUIREMENTS:

Dimensions and configurations: See figure 1. This specification sheet describes the socket side of a rectangular, plastic shell connector. This connector uses reverse gender contact, i.e., the live pin recessed in the insulator with the static socket protruding from a shrouded interface.

Material and finish:

Shell: The requirements for shell materials shall be in accordance with MIL-DTL-83513.

Contact identification: See figure 2 for contact layout identification.

Contact connection: The pin contact (being recessed in the insulator) is normally connected to the live side of the circuit.

Current rating, maximum: 3 amperes per contact.

Mounting and mating hardware: Mounting hardware is to be ordered separately (see MIL-DTL-83513/5, configurations A and B).

Mating plug: Shall conform to MIL-DTL-83513/6 or MIL-DTL-83513/8.

Example of Part or Identifying Number (PIN):

M83513/09 - A 01 1/

M83513/09: Specification sheet number

A: Insert arrangement (see figure 2)

01: Wire type

A = 9
B = 15
C = 21
D = 25
E = 31
F = 37
G = 51

01 = M22759/11-26-9, 18 inches long 2/ /
02 = M22759/11-26-9, 36 inches long 2/ /
03 = M22759/11-26-( ), 18 inches long 2/ 3/
04 = M22759/11-26-( ), 36 inches long 2/ 3/
05 = A-A-59551, type S, 25 AWG, except shall be gold plated 10 microinches; 0.5 inch long 4/
06 = A-A-59551, type S, 25 AWG, except shall be gold plated 10 microinches; 1.0 inch long 4/
07 = A-A-59551, type S, 25 AWG, tin plated 100 microinches; 0.5 inch long 4/
08 = A-A-59551, type S, 25 AWG, tin plated 100 microinches; 1.0 inch long 4/
09 = M22759/33-26-9, 18 inches long 2/
10 = M22759/33-26-9, 36 inches long 2/
11 = M22759/33-26-( ), 18 inches long 2/ 3/
12 = M22759/33-26-( ), 36 inches long 2/ 3/
13 = M22759/11-26-9, 72 inches long 2/
14 = M22759/11-26-( ), 72 inches long 2/
15 = M22759/33-26-9, 72 inches long 2/
16 = M22759/33-26-( ), 72 inches long 2/

See footnotes on next page.

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1/ These connectors have leads attached of which wire type, wire size, wire color, and lead length are specified in PIN.
2/ Lead length tolerance +1.0, -0.0 inch.
3/ Color coding in accordance with MIL-STD-681, system I, except the appropriate color code shall be the individual contact cavity number minus 1 for contact positions 1 through 10 only, 11 through 20, etc. For example:

Pin number	Identification
4	3
9	8
21	0
31	0
47	6

4/ Lead length tolerance +0.2, -0.0 inch.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AT, MI
Navy - AS, CG, MC, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project: 5935-4310-009)

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